

Title

Rehabilitation Session Characteristics and Activity Levels in Post-Stroke Inpatient Care

Abstract

We characterized rehabilitation intensity during inpatient physical therapy sessions across Canadian rehabilitation centers. One hundred sixty-six stroke survivors (mean age 69 years, 42% female) admitted for inpatient rehabilitation were assessed during a single midpoint rehabilitation session using wearable monitors. Mean aerobic exercise time (defined as >40% of individual heart rate reserve) was 11 minutes (SD 12) per session, with wide variability. Mean step count was 1,030 steps (SD 620). Aerobic minutes and step count showed negligible correlation ($R^2 = 0.002$), suggesting these represent relatively independent dimensions of activity. Participants achieving ≥ 20 aerobic minutes during the session (22% of sample) were more likely to be male, have lower baseline walking distance, and have subcortical stroke location. No significant demographic differences were observed in mean step count. Current practice falls substantially short of guideline recommendations for aerobic intensity. Session content appeared to emphasize stepping activities over cardiovascular conditioning. These findings highlight a potential implementation gap between recommended stroke rehabilitation protocols and typical clinical practice.